Rotating Base

MCU
MCU.SchDoc

RF
RadioRF.SchDoc

RF1 — RF1

MOTOR_INT — MOTOR_INT
LED_CTRL — LED_CTRL
TOP_UI — TOP_UI

SENS
Sensor_Suite.SchDoc

ACCEL_INT1 — ACCEL_INT1
I2C_SCL — I2C_SCL
I2C_SDA — I2C_SDA

IR_LED_EN — IR_LED_EN
ROT_INDEX — ROT_INDEX
INDEX_RAW — INDEX_RAW

MTR
Motor_Control.SchDoc

MOTOR_INT — MOTOR_INT

REG
Power_Distribution.SchDoc

Rotating Vertical

LED
LED_Matrix.SchDoc

LED_CTRL — LED_CTRL
TOP_UI — TOP_UI

Top Interface

SIDE
Side_UI.SchDoc

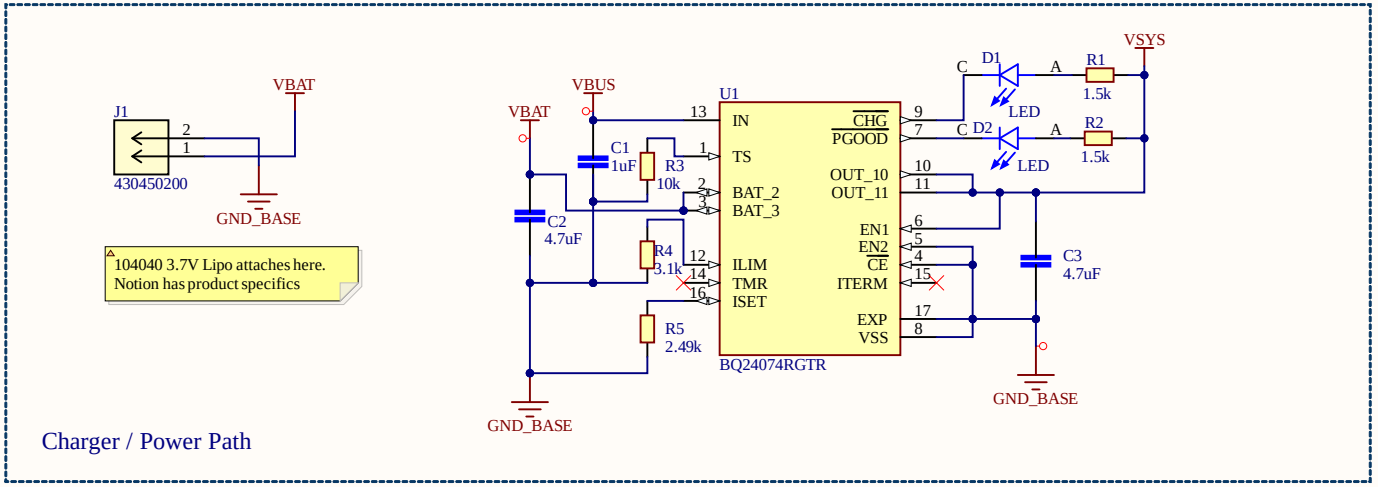
TOP_UI — TOP_UI

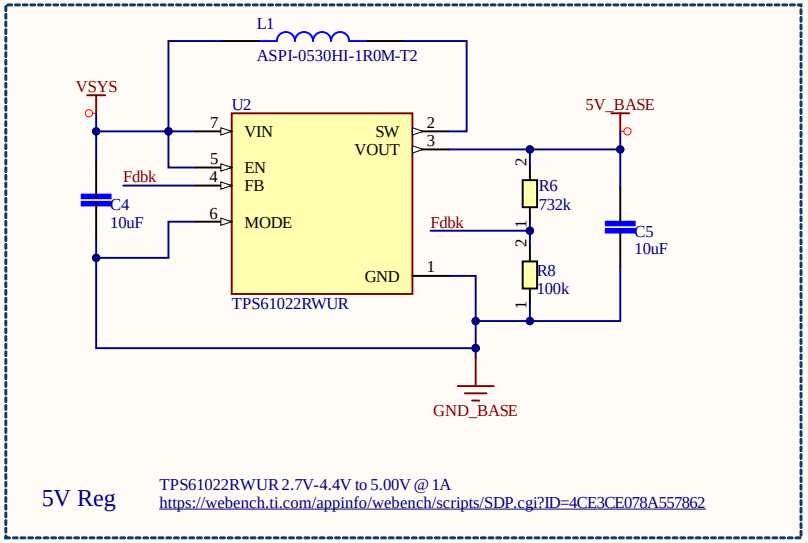
Stationary Base

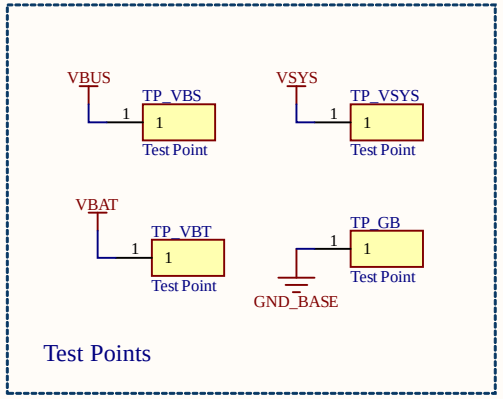
USB
USB_Debug.SchDoc

CHG
Power_Charging.SchDoc

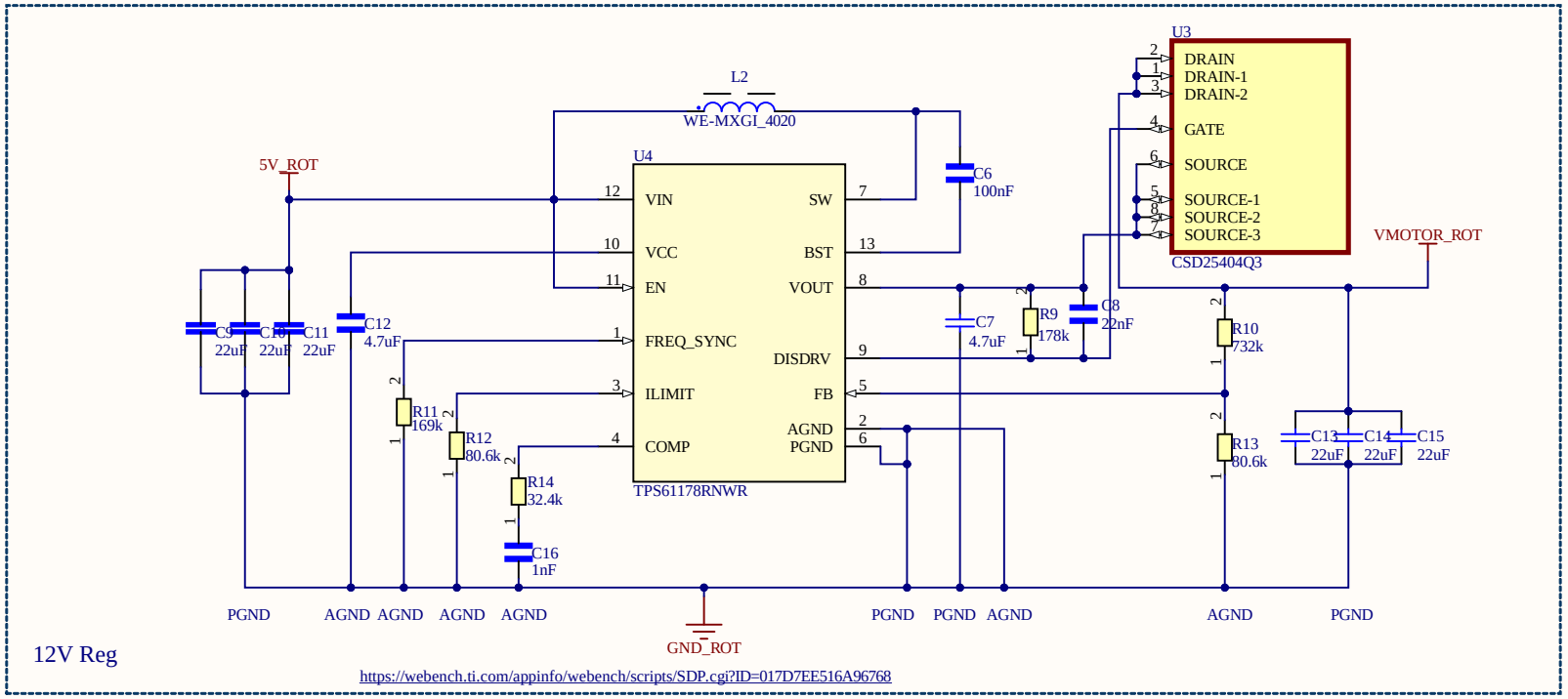
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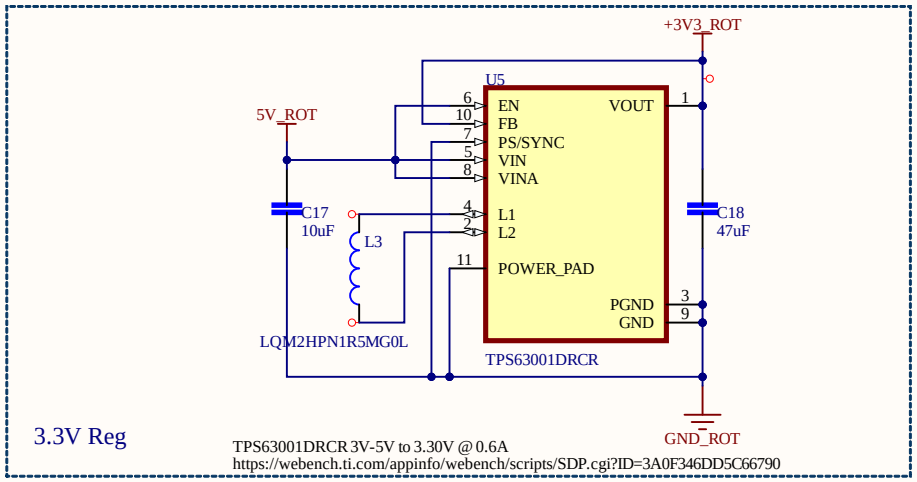


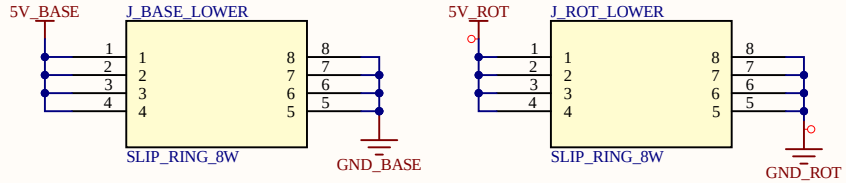




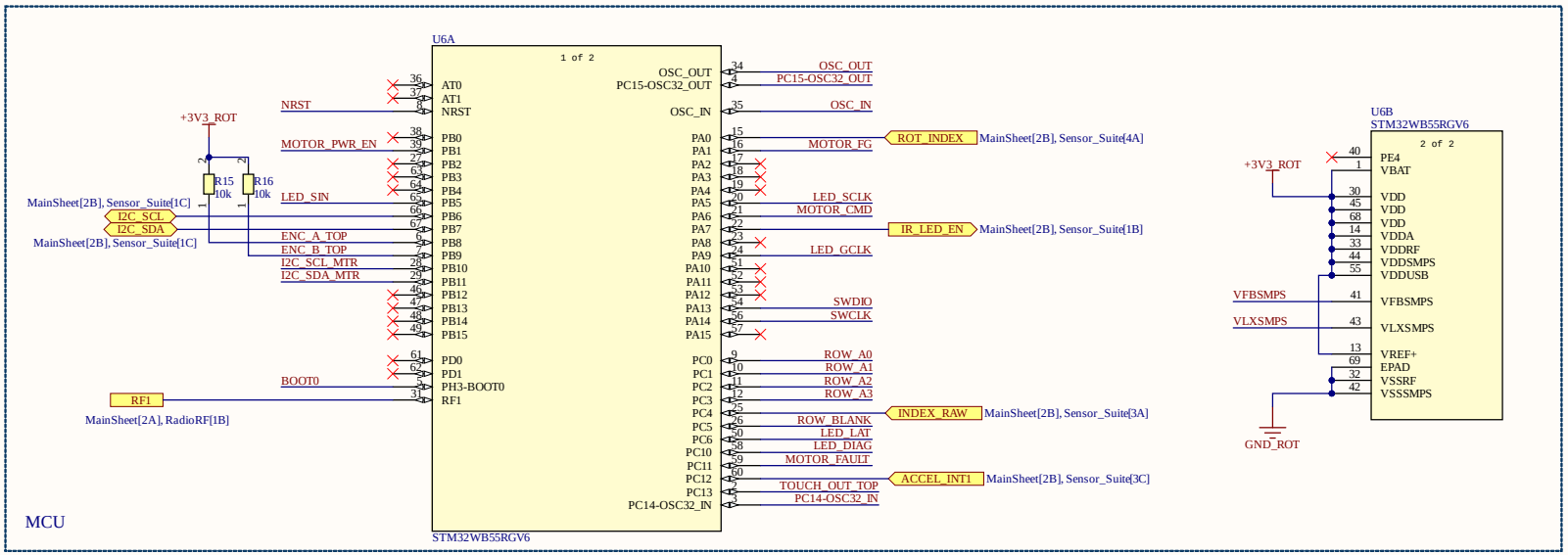
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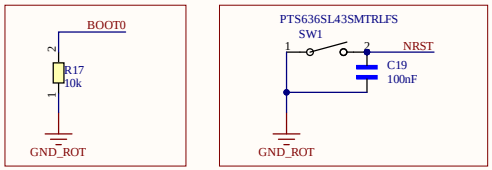


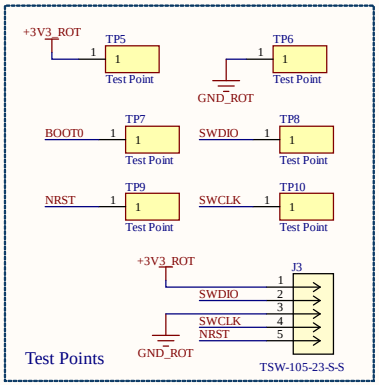


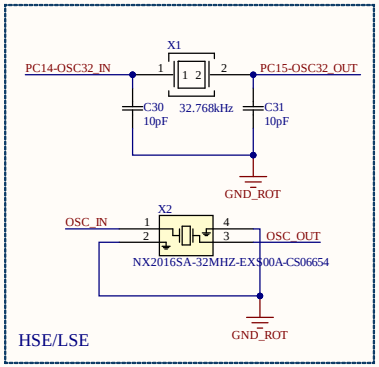


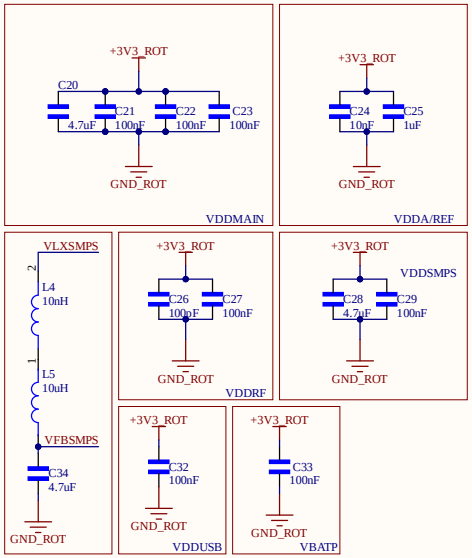
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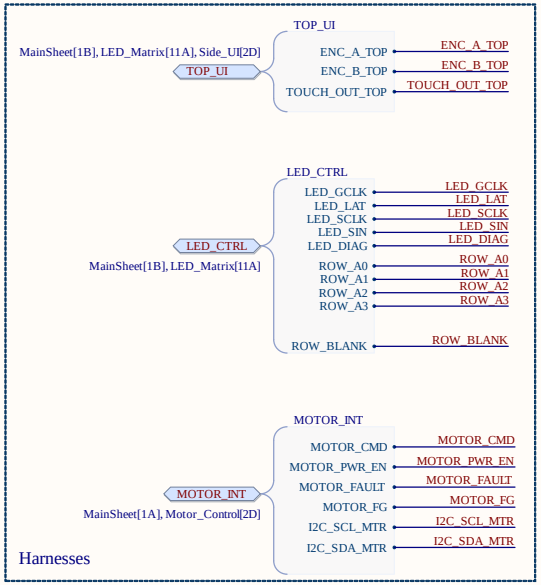




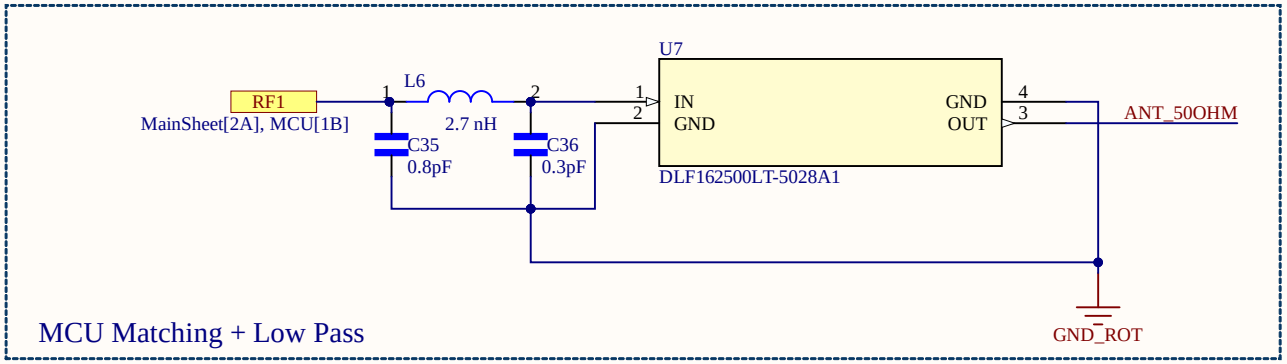


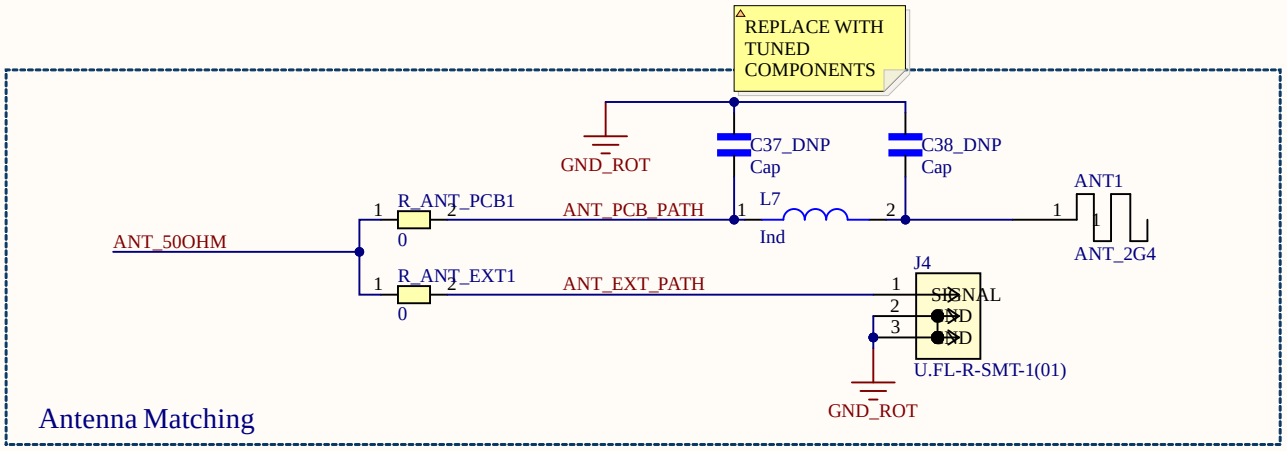




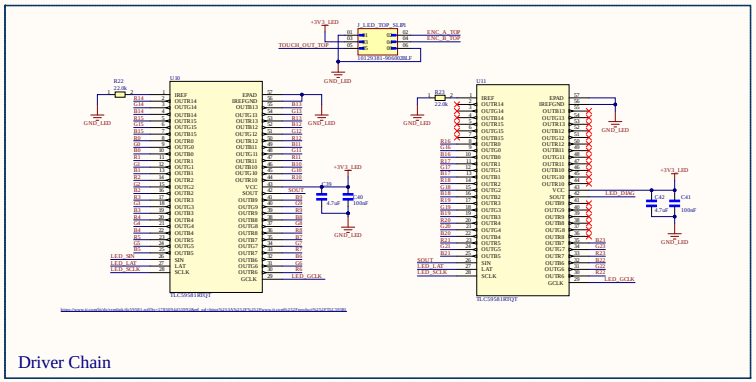


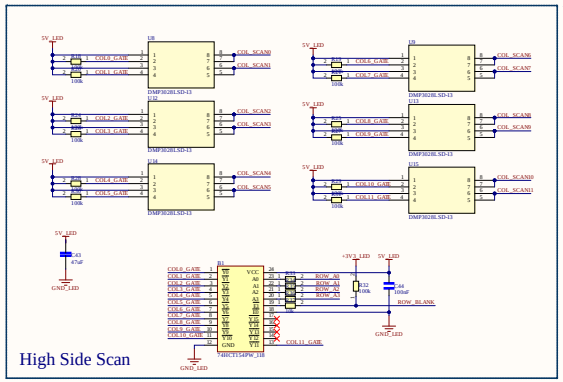
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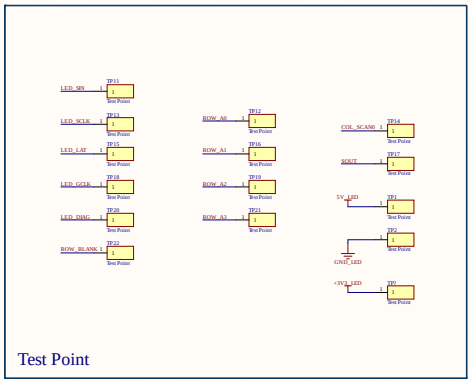


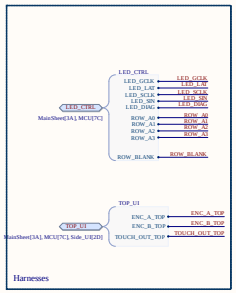


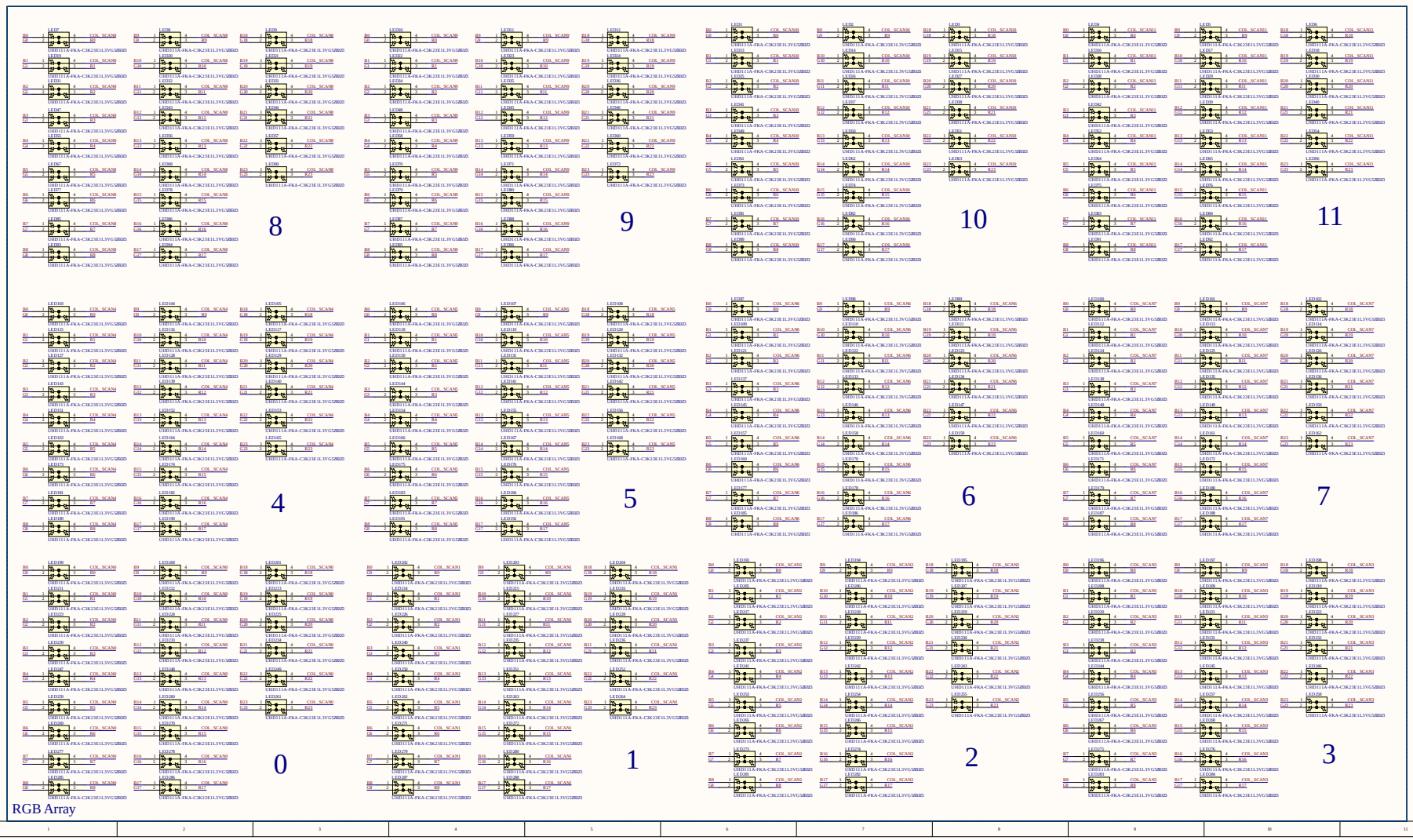
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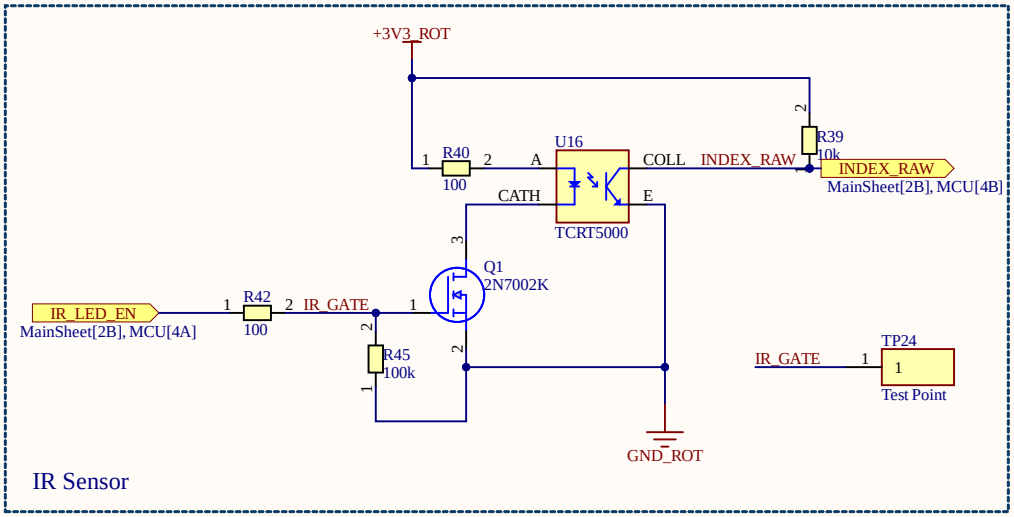
ARCHITECTURE PLAN

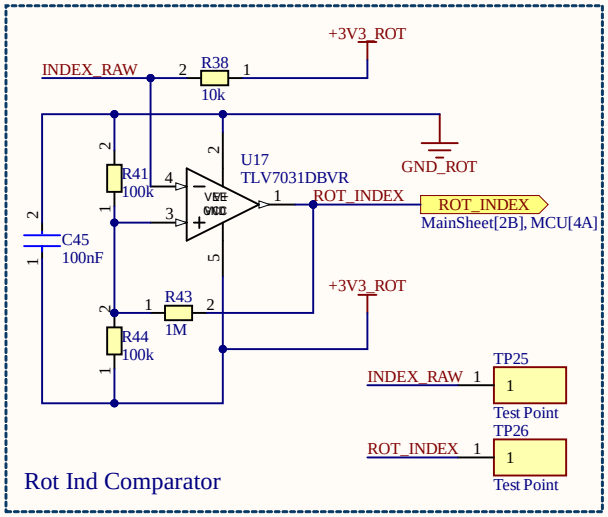
12 x 24 = 288 RGB LEDS
Scanning will use one common anode column active at a time 1/12 multiplexing

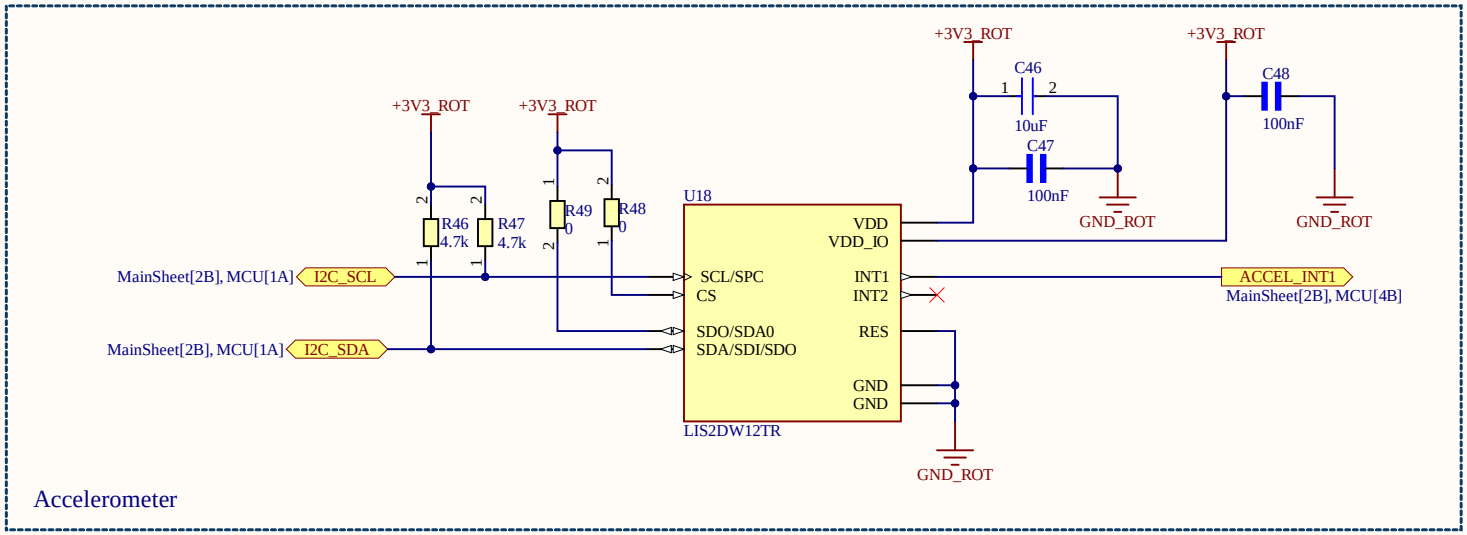
Driven with 2 x TLC59581RTQR
- 72 / 96 outputs used
- 24 outputs reserved

POWER
+3V3 for driver and decoder logic
VLED_5V for LED anode power

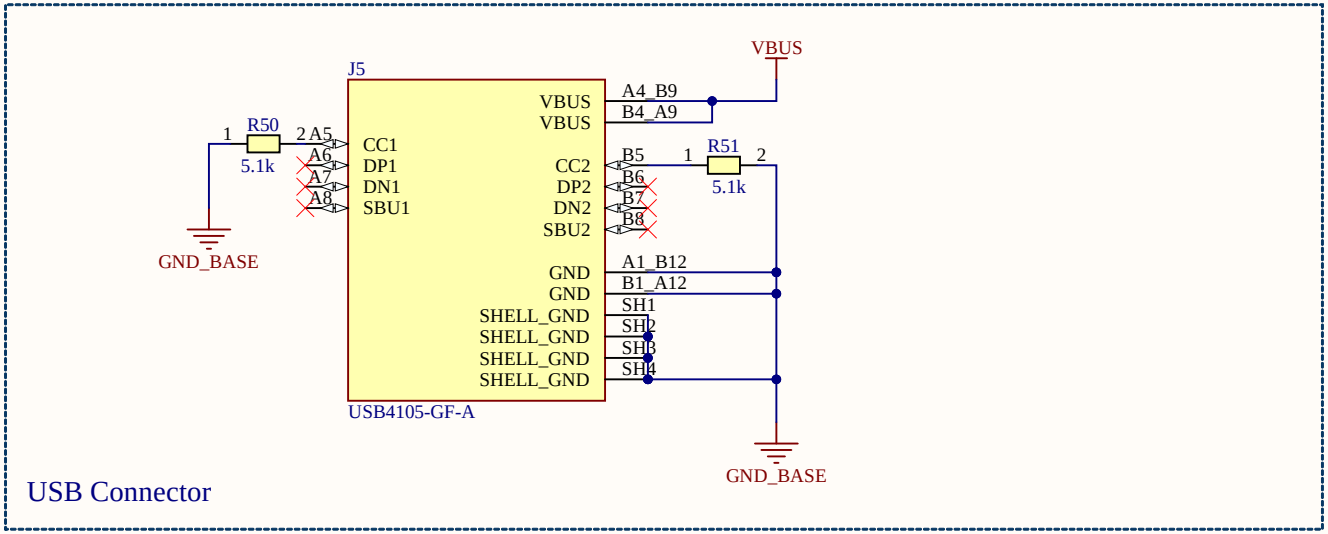
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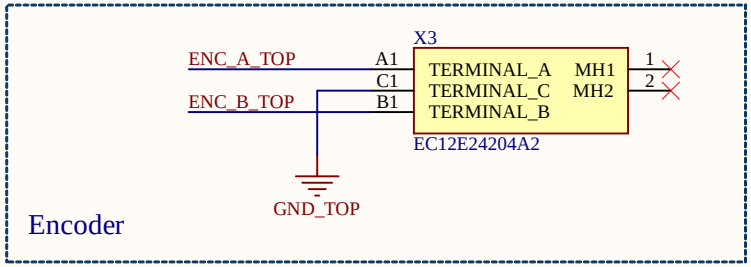


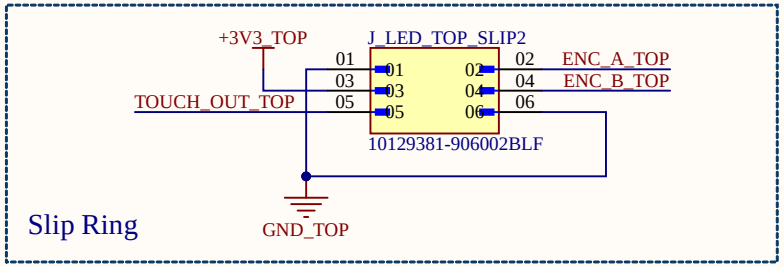


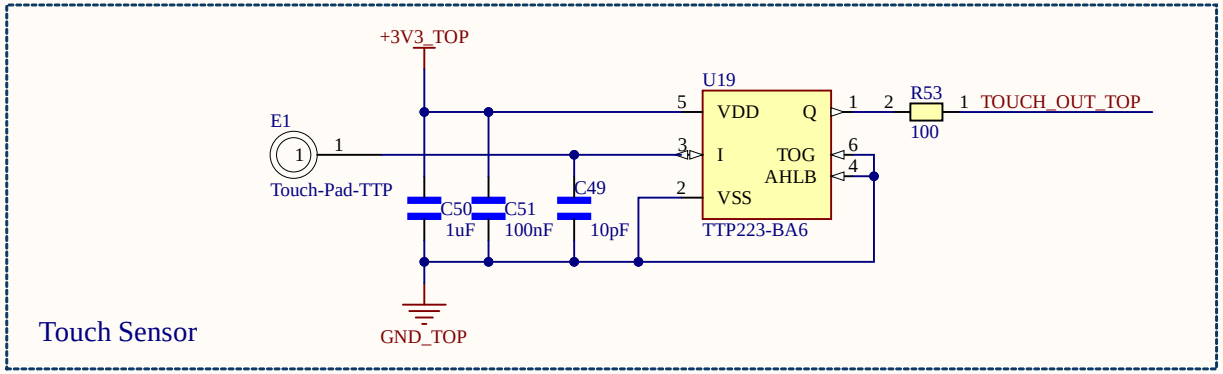
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Size A4	Number	Revision
Date: 9/12/2026		Sheet of
File: C:\Users\..\Sensor_Suite.SchDoc		Drawn By:

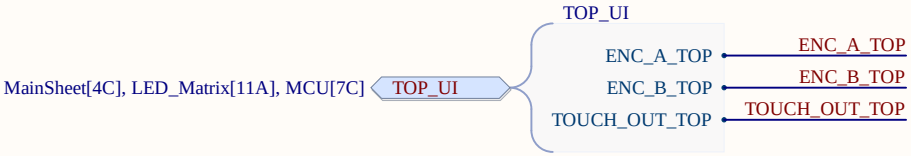


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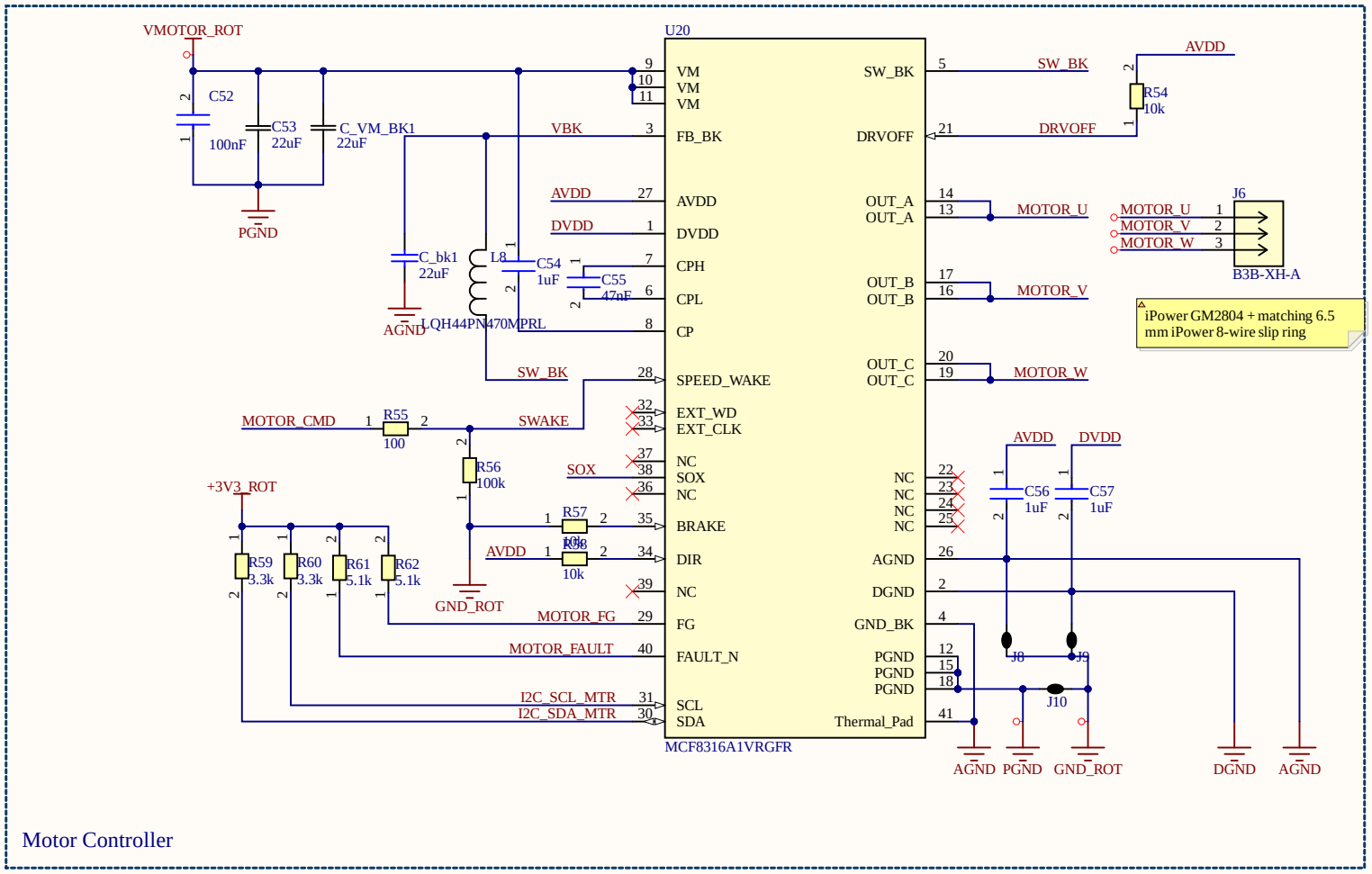


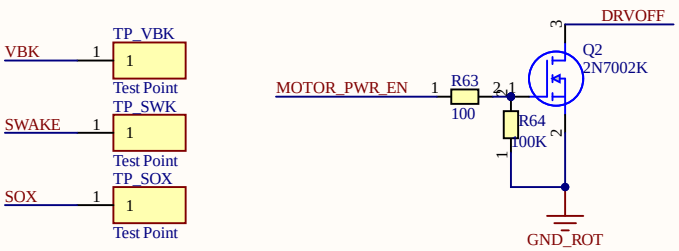


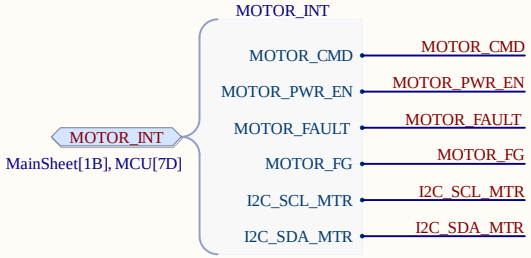




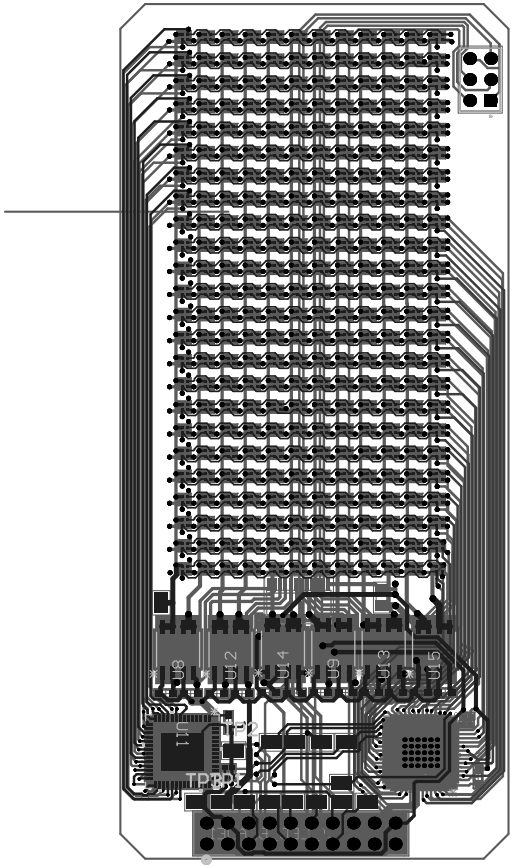
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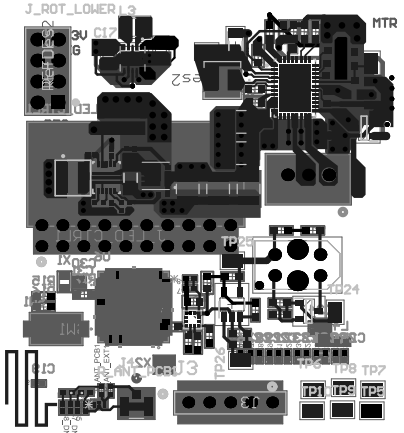


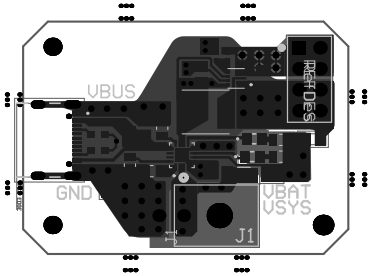




Title		
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Date: 9/12/2026		Sheet of
File: C:\Users\..\Motor_Control.SchDoc		Drawn By:




J_ROT_LOWER_L3
MTR
3V
G
TP6
TP8
TP7
TP1
TP3
TP2


VBUS
GND
VBAT
VSYS
J1

